

Computing Publications with Major Personal Contributions

Oliver Gutsche

June 01, 2026

1. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

2. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

3. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels

Kwok, Ka Hei Martin et al.

[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

4. The U.S. CMS HL-LHC R D Strategic Plan

Gutsche, Oliver et al.

[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

5. A Ceph S3 Object Data Store for HEP

Smith, Nick et al.

[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

6. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

7. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

[arXiv:2306.13567](#) | 2023-06-23

8. Automated Network Services for Exascale Data Movement

Balcas, Justas et al.

[EPJ Web Conf. 295 \(2024\) 01009](#) | [DOI:10.1051/epjconf/202429501009](#) | 2024

9. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.

[arXiv:2302.01317](#) | 2023-02-02

10. The Future of High Energy Physics Software and Computing

Elvira, V. Daniel et al.

[arXiv:2210.05822](#) | [DOI:10.2172/1898754](#) | 2022-10-11

11. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization

Cerati, G. et al.

[arXiv:2209.07356](#) | 2022-09-15

12. Portability: A Necessary Approach for Future Scientific Software

Bhattacharya, Meghna et al.

[arXiv:2203.09945](#) | 2022-03-15

13. Coffea: Columnar Object Framework For Effective Analysis

Smith, Nicholas et al.

[EPJ Web Conf. 245 \(2020\) 06012](#) | [arXiv:2008.12712](#) | [DOI:10.1051/epjconf/202024506012](#) | 2020-08-28

14. Using Big Data Technologies for HEP Analysis

Cremonesi, Matteo et al.

[EPJ Web Conf. 214 \(2019\) 06030](#) | [arXiv:1901.07143](#) | [DOI:10.1051/epjconf/201921406030](#) | 2019-01-21

15. HEP Software Foundation Community White Paper Working Group – Data Organization, Management and Access (DOMA)

Berzano, Dario et al.

[arXiv:1812.00761](#) | 2018-11-30

16. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation

Bauerdick, Lothar et al.

[arXiv:1804.03983](#) | 2018-04-09

17. A Roadmap for HEP Software and Computing R D for the 2020s

Albrecht, Johannes et al.

[Comput.Softw.Big Sci. 3 \(2019\) 7](#) | [arXiv:1712.06982](#) | [DOI:10.1007/s41781-018-0018-8](#) | 2017-12-18